Conditional Deep Lévy Models for Exotic Derivatives: History-Aware Path Generation and P–Q Payoff Diagnostics

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Abstract

We develop and audit a history-aware financial path generator based on Denoising Lévy Probabilistic Models (DLPMs) for conditional equity-index path generation. The model combines symmetric α-stable diffusion noise with a conditional U-Net observing the contract state, 60- and 252-day return histories, trend, drawdown, volatility paths, country, and index identity. A chronological protocol evaluates a frozen model on 6,824 untouched windows from eight Chinese and U.S. equity indices. The final generator attains terminal CRPS 0.0485, path energy score 0.3368, a generated-to-realized volatility ratio of 0.882, and aggregate signed terminal-median bias of -0.46 percentage points. It improves terminal CRPS over contract-only DLPM and Gaussian diffusion controls, while an unconditional historical block bootstrap remains competitive on pooled marginal metrics. Against a stronger state-matched empirical control that selects same-index, same-tenor training windows using pre-window volatility, trend, and drawdown, DLPM lowers terminal CRPS from 0.0633 to 0.0485 (23.3%); for this state-matched CRPS comparison, joint calendar-block confidence intervals remain below zero for block lengths from 20 to 150 days.

We then use the learned physical-measure distribution in a transparent P–Q payoff diagnostic. The Q side is an independently fitted Student-t GARCH(1,1) benchmark with capped, unit-variance innovations and a numerical log-mgf correction that enforces the stated discrete martingale restriction. At the pre-specified $g = 0.05$ dealer-spread setting and a $1\%S_0$ materiality threshold, the aligned, unconditional P–Q diagnostic is 0.91% of initial spot for vanilla calls and 0.36% for Asian calls. Complementary drift-neutral and initial-delta diagnostics decompose this payoff difference into direction-sensitive and residual components. Physical direction is an important component of the raw gap; under the one-time RN-Q Delta diagnostic, the Vanilla residual is 0.33% of spot while the Asian residual is close to zero. The P–Q framework is a belief-based payoff diagnostic, distinct from an option-surface-calibrated pricing engine.

Keywords: denoising Lévy probabilistic model; conditional diffusion; exotic options; physical measure; risk-neutral benchmark; GARCH; scenario generation; structured products.

JEL classification: C45, C53, G12, G13, G17.

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1 Introduction

Financial institutions routinely value claims whose cash flows depend not only on the terminal asset price but also on averages, extrema, observation schedules, and the order in which market states are visited. Asian and lookback options are canonical examples; accumulators and autocallable notes combine path dependence with leverage, early termination, and nonlinear coupon rules. Their risk cannot be summarized by a point forecast. What matters is a conditional distribution over entire future paths.

Classical Monte Carlo valuation begins by specifying a stochastic process and then averaging discounted payoffs. This workflow is transparent, but its output inherits the restrictions of the assumed process. Geometric Brownian motion imposes constant volatility and Gaussian increments. Local and stochastic-volatility models are more flexible, yet they remain parsimonious parametric representations of a market in which heavy tails, volatility clustering, drawdowns, trend reversals, and regime-dependent dependence are empirically prominent (Cont, 2001). These restrictions are especially consequential for nonlinear claims: two models can agree on one-step volatility and still disagree sharply on an Asian average, a running maximum, or the sequence of states that activates a structured payoff.

Generative models offer a complementary route. Rather than committing to a small set of diffusion coefficients, one can learn a conditional law of future returns directly from historical paths. Denoising diffusion models are attractive because they provide stable likelihood-inspired training, broad support, and a natural mechanism for conditional scenario generation (Ho et al., 2020; Rasul et al., 2021). A Gaussian forward process, however, still builds light-tailed noise into the generative mechanism. For financial returns, this is not an innocuous computational choice. Large moves are rare but economically decisive, and the reverse process must represent them without either suppressing tail mass or producing numerically uncontrolled paths.

This paper develops a heavy-tailed alternative based on the Denoising Lévy Probabilistic Model of Shariatian et al. (2024). The DLPM replaces Gaussian noise with symmetric

α-stable innovations while retaining a tractable variance-mixture representation. We combine this mechanism with a history-aware conditional network and a finance-aware auxiliary objective. The conditioning information is deliberately richer than the contract state: it includes 60- and 252-day return sequences, rolling volatility, trend, drawdown, downside risk, reversal indicators, country, and underlying index. The resulting model is a conditional physical-measure path generator. It asks: given what was observable at the start of a contract window, what distribution of future index paths is plausible?

The second contribution is economic. A path generator should not be judged only by a pooled return histogram. Small distributional differences can matter greatly for nonlinear payoffs, while visually different paths can imply nearly identical vanilla values. We therefore embed the frozen DLPM in a P–Q quoting experiment. The P side represents the learned real-world belief. The Q side is an independently fitted Student-t GARCH(1,1) benchmark whose drift is martingale-corrected under the simulated innovation law. The two models value the same payoff; a trade is opened only when their valuation gap clears both the dealer spread and a pre-specified execution threshold; the held-out realized path determines ex-post P&L. This construction converts path differences into payoff-sensitive evidence without using downstream profits to train or select the generator.

The distinction between P and Q is central. Historical conditional generation and arbitrage-free pricing are related but different tasks. The DLPM is not trained on option prices and is not subjected to a stochastic discount factor; it therefore represents a physical-measure belief rather than a market-implied risk-neutral law. Conversely, fitting a GARCH model to historical returns does not by itself create a Q measure. Our benchmark explicitly sets the conditional drift through a numerical log-mgf adjustment and verifies the discounted martingale property. It remains a transparent benchmark Q, not a reconstruction of the unique market-implied measure.

The empirical design is unusually broad for a financial diffusion study. The sample contains CSI 300, CSI 500, CSI 1000, SSE Composite, S&P 500, NASDAQ, Dow Jones, and

Russell 2000. The final protocol uses 59,240 training windows, 3,084 chronological validation windows, and 6,824 untouched test windows. The validation period selects among EMA checkpoints; the test period is used once for the final path and economic audit. Because the sliding windows overlap and the indices share calendar shocks, we do not treat 6,824 windows as independent observations. Economic uncertainty is estimated with joint calendar-block bootstrap resampling at four block lengths.

The final results establish a broad conditional-generation capability. The aggregate signed terminal-mean drift error is -0.27 percentage points and the generated-to-realized volatility ratio is 0.882. Conditional fans anchor at the observed index level, change scale across markets, and respond to the pre-window state. Crucially, it lowers terminal CRPS by 23.3% against a same-index, same-tenor state-matched empirical control, with dependence-robust confidence intervals excluding zero across all reported block lengths. The aligned P–Q audit then converts these path differences into payoff-sensitive diagnostics, with unconditional values of 0.91% for vanilla and 0.36% for Asian calls at the primary dealer-spread setting and the pre-specified $1\%S_0$ materiality threshold.

The paper makes five contributions:

1. It provides a full theoretical and computational formulation of conditional Gaussian diffusion, finance-aware diffusion objectives, and heavy-tailed DLPM training and sampling for financial paths.

2. It introduces a history-aware condition representation that combines raw pre-start return sequences with interpretable trend, volatility, drawdown, reversal, country, and index features.

3. It evaluates one global generator on eight equity indices under a chronological train–validation–test protocol with target-end purging and no payoff-based checkpoint selection.

4. It constructs and audits a martingale-corrected Student-t GARCH RN-Q benchmark,

including post-truncation unit-variance normalization.

5. It links distributional path diagnostics to a payoff-sensitive P–Q experiment with normalized results, spread sensitivity, and joint calendar-block bootstrap uncertainty.

Figure 1 summarizes the complete system. The upper branch learns the P-side conditional distribution; the lower branch constructs the independent RN-Q benchmark; the final branch evaluates payoff beliefs against held-out realizations.

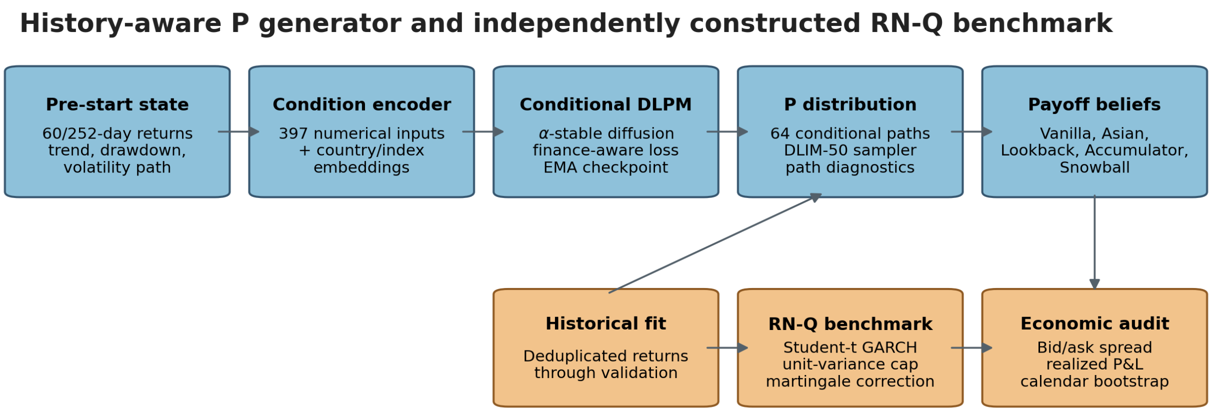

Figure 1: End-to-end research design. The DLPM learns conditional physical-measure paths from pre-start information. The Student-t GARCH benchmark is fitted independently and martingale-corrected. Option payoffs and the P–Q game are downstream audits and do not enter model training or checkpoint selection.

The remainder of the paper proceeds as follows. Section 2 reviews exotic-option valuation, financial generative models, and alternative time-series architectures. Section 3 presents the cross-market data and temporal protocol. Section 4 develops the Gaussian and Lévy diffusion theory. Section 5 reports checkpoint selection and path quality. Section 6 constructs the RN-Q benchmark and presents the economic results. Section 7 consolidates the limitations and scope of the study, and Section 8 concludes.

2 Related Literature

2.1 Pricing structured products and exotic options

Industry practice relies on four methodologies (Deng et al., 2014): simulation, which averages discounted payoffs over generated price paths and suits strongly path-dependent products; numerical integration, when payoff and density admit closed forms; decomposition into bonds, vanilla options, and simpler exotics; and partial differential equation methods, which capture continuous state dynamics at high solution cost. The Black–Scholes model, despite its centrality for vanilla options, handles multi-layered structures poorly given its risk-neutrality assumption and single volatility parameter.

A long literature refines these tools: Monte Carlo pricing from Boyle (1977), jump-diffusion dynamics from Merton (1976), Brownian-bridge quasi-Monte Carlo constructions (Lin and Wang, 2008), ant colony optimization (Kumar et al., 2009), and tensor-network acceleration of lattice pricing (van Damme et al., 2025). Machine-learning approaches include LSTM option pricing (Pimentel et al., 2025), generative adversarial models for arbitrage-free implied volatility surfaces (Vuletić and Cont, 2025), and risk-neutral pricing via GANs (Choi et al., 2025).

2.2 Diffusion models for time series

Denoising diffusion probabilistic models were established by Ho et al. (2020), refined by the cosine schedule of Nichol and Dhariwal (2021), and accelerated by the deterministic DDIM sampler of Song et al. (2021). TimeGrad (Rasul et al., 2021) pioneered conditional diffusion for probabilistic time-series forecasting; DiffSTOCK (Daiya et al., 2024) applies relational diffusion to equity prediction; Meijer and Chen (2024) survey the area.

Diffusion models have recently become the leading generative approach for financial time series. Takahashi and Mizuno (2025) show that denoising diffusion models reproduce the stylized facts of asset returns—heavy tails, volatility clustering, and dependence structure—

more faithfully than GANs and variational autoencoders, without mode collapse; subsequent work extends the framework to conditional multivariate generation with asset-specific and systematic covariates. Yet the same literature emphasizes an open problem: no generator has fully captured all stylized facts simultaneously, with tail behavior proving the most stubborn. This gap motivates our heavy-tailed model class.

Classical diffusion is Gaussian-driven. For heavy-tailed data, Shariatian et al. (2024) propose Denoising Lévy Probabilistic Models (DLPMs), whose original formulation replaces Gaussian increments with isotropic α-stable noise, $\alpha \in (1, 2]$ (Samorodnitsky and Taqqu, 1994). We retain the original acronym and use "conditional deep Lévy model" only as a descriptive label for our financial adaptation, not as a renaming of the cited method. Our financial-path implementation preserves the same stable-law mechanism but uses *coordinate-wise* stable scale mixing across ordered return coordinates, instead of one path-level isotropic scale draw. Exploiting the variance-mixing representation of stable laws, this retains DDPM's computational structure with minimal changes while improving tail coverage; $\alpha = 2$ recovers the Gaussian DDPM exactly. Given the well-documented heavy tails of asset returns (Cont, 2001), DLPM is a natural generator for financial paths; to our knowledge this paper is the first to evaluate it in a derivatives-valuation setting.

2.3 Alternative model families

Financial market generators provide a closely related line of work. Wiese et al. (2020) use temporal-convolutional GANs to reproduce long-range dependence and volatility clustering, while Kondratyev and Schwarz (2019) develop a restricted-Boltzmann-machine market generator for dependence-aware scenario sampling. Signature methods replace a learned discriminator with path-signature distances: Liao et al. (2024) learn conditional time-series laws through a Sig-Wasserstein objective. For derivative applications, Cuchiero et al. (2020) calibrate neural local-stochastic-volatility models to option prices, and Buehler et al. (2019) learn hedging policies under frictions. These studies establish complementary benchmarks for scenario fidelity, risk-neutral calibration, and downstream hedging. Our setting differs by learning a physical-measure, history-conditioned path law with an explicit heavy-tailed diffusion mechanism and then auditing its payoff implications against an independently constructed RN-Q benchmark.

Two further families compete in this space. *Neural stochastic differential equations* parameterize the drift and diffusion of an SDE by neural networks, trained by likelihood or adversarial criteria (Gierjatowicz et al., 2020); imposing no-arbitrage constraints yields market models suitable for risk-neutral applications (Cohen et al., 2023). *Time-series foundation models* such as Chronos (Ansari et al., 2024) and TimesFM (Das et al., 2024) are transformers pretrained on billions of observations that produce zero-shot probabilistic forecasts; however, generic foundation models tend to underperform domain-specific architectures on financial data, and their uncertainty calibration remains an open concern. Because both families are natural alternatives to diffusion generators, our empirical study includes a conditional neural-SDE baseline trained on identical data (Section 5), positioning the diffusion results against the classical SDE paradigm rather than against Gaussian Monte Carlo alone.

3 Data and Experimental Protocol

3.1 Markets, dates, and window construction

The data set covers eight broad equity indices in two large markets:

CSI 300, CSI 500, CSI 1000, SSE Composite, S&P 500, NASDAQ, Dow Jones, and Russell 2000.

The raw collection spans 1 January 2014 through 1 August 2025. Chinese index levels are obtained from AKShare (tickers 000300, 000905, 000852, and 000001), and U.S. adjusted index series from Yahoo Finance (`ĜSPC`, `ÎXIC`, `D̂JI`, and `R̂UT`). The series are price-index or provider-adjusted index levels, not total-return indices. Chinese rates use SHIBOR 1-, 3-, 6-, and 12-month tenors; U.S. rates use FRED DGS1MO, DGS3MO, DGS6MO, and DGS1. Rates are matched to contract tenor, converted to annual decimals, and forward-filled on the local business-day calendar.

A 252-trading-day pre-start history requirement and complete-target filtering determine the usable range. Windows have calendar maturities of 30, 90, 180, and 365 days and record initial price, realized future path, historical volatility, risk-free rate, country, and index identity.

Table 1: Chronological sample construction

Split	Windows	Earliest start	Latest start	Latest target end
Training	59,240	2015-01-02	2022-11-11	2022-12-13
Validation	3,084	2023-01-03	2023-10-02	2023-11-01
Test	6,824	2024-01-02	2025-06-06	2025-07-06

Counts are after target-end purging. Target end is reconstructed from the window start, contractual calendar horizon, and stored realized path, and was checked against the realized-price archive. The test set contains 827 windows for each Chinese index and 879 for each U.S. index.

The split is chronological and materialized before training. For each underlying, a target end date is reconstructed from its ordered trading calendar and stored path length. A window in the training or validation split is retained only if its estimated target end precedes the next split boundary by at least 20 calendar days. The implementation labels this operation as a purge/embargo; the actual code uses a 20-calendar-day buffer rather than claiming 20 exchange-specific trading days. This distinction is documented because it matters for exact replication. The resulting maximum target dates in Table 1 leave visible separation from the following split's start.

3.2 Returns, scales, and historical state

For index level S_t, daily log return is

$$r_t = \log(S_t/S_{t-1}), \tag{1}$$

and annualized historical volatility over n observations is

$$\sigma_{\text{ann}} = \sqrt{252}\left(\frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2\right)^{1/2}. \tag{2}$$

The model represents the target in scaled log-return space,

$$x_{0,t} = r_t/s_{\text{vol}}, \qquad s_{\text{vol}} = 0.09, \tag{3}$$

and reconstructs levels by exact anchoring at S_0:

$$\widehat{S}_t = S_0 \exp\left(s_{\text{vol}}\sum_{j=1}^{t}\widehat{x}_{0,j}\right). \tag{4}$$

Equation (4) guarantees that every generated path starts at the observed contract level. Variable-tenor windows are padded to length 252 and accompanied by a validity mask; padded entries do not contribute to the training objective. Price observations with missing dates or levels are dropped rather than interpolated. Exchange holidays and suspensions therefore shorten the stored realized path; payoff evaluation uses the recorded `actual_trading_days` and never fabricates unavailable prices.

The numerical condition contains 397 entries. It combines contemporaneous contract variables with strictly pre-start information: raw 60- and 252-day return sequences, a 60-day volatility path, 20/60/252-day trends, current and maximum drawdown, realized and downside volatility, recent reversal, momentum acceleration, and ratios between short- and long-horizon volatility. Two categorical identifiers encode country and underlying. Learned embeddings allow the global model to share common financial structure while retaining market-specific behavior.

The raw sequences and summary variables play different roles. Sequences retain local temporal ordering; summaries expose low-frequency state variables that a convolutional encoder might otherwise learn inefficiently. This combination is particularly useful at long horizons, where a network must distinguish a high-volatility rebound from a low-volatility trend even when their one-day return distributions overlap.

3.3 Dependence and effective sample size

Sliding windows are intentionally dense because they expose the model to many conditional states. They do not create an equal number of independent experiments. Adjacent windows share both historical context and forward targets, and indices within the same country share market-wide shocks. Accordingly, the paper never interprets 6,824 as an effective independent sample size. Path metrics are descriptive averages over the complete held-out set. Economic uncertainty is estimated by sampling common calendar blocks for all assets, preserving both serial overlap and cross-index synchronization. We report 20- and 60-day blocks as the primary dependence-robust inference checks, 90-, 120-, and 150-day blocks as medium-horizon sensitivities, and 180- and 252-day blocks as long-dependence sensitivities, each with 2,000 replications. The latter two have only three nonempty calendar blocks and are reported for transparency rather than treated as decisive inference.

3.4 Validation and information discipline

The network is optimized for 16,000 steps. EMA checkpoints at 4,000, 8,000, 12,000, and 16,000 steps are evaluated on the validation period with fixed random seeds and the deployed 50-step sampler. Selection uses the pre-specified validation score

$$\mathcal{S}_{\mathrm{val}} = |\widehat{C}_{0.9} - 0.9| + 0.5\,\mathrm{CRPS} + 0.25\,\mathrm{Energy} + 0.5|\log(\widehat{\sigma}/\sigma)| + 0.25|\Delta\mathrm{ACF}| + 0.1|\Delta\mathrm{Median}|. \quad (5)$$

The component diagnostics additionally include terminal quantiles, tail behavior, autocorrelation, and drawdown. No option payoff, P–Q profit, or test observation enters checkpoint selection. EMA-12000 is frozen before the 2024–2025 test audit.

4 Methodology

4.1 Gaussian conditional diffusion

The forward process of a DDPM is a Markov chain that gradually corrupts the data $x_0 \in \mathbb{R}^{B\times C\times L}$ with Gaussian noise,

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(x_t;\ \sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I\Big), \tag{6}$$

under the cosine schedule β_t of Nichol and Dhariwal (2021) with $T = 1000$ steps in the final clean-split deployment. With $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i\le t} \alpha_i$, the marginal admits the reparameterization

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I). \tag{7}$$

The reverse process approximates $q(x_{t-1} \mid x_t, x_0)$ by a Gaussian transition conditioned on contract features c,

$$p_\theta(x_{t-1} \mid x_t, c) = \mathcal{N}\Big(x_{t-1};\ \mu_\theta(x_t, t, c),\ \tilde{\beta}_t I\Big), \tag{8}$$

where the conditional U-Net (Appendix A) outputs $\hat{x}_0 = f_\theta(x_t, t, c)$ and the posterior mean and variance follow from Bayes' rule for the Gaussian chain:

$$\tilde{\mu}_t(x_t, \hat{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\, \beta_t}{1-\bar{\alpha}_t}\, \hat{x}_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t, \qquad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\, \beta_t. \tag{9}$$

For fast generation we use the deterministic DDIM sampler (Song et al., 2021): for a subsequence $\{\tau_i\} \subset \{1, \ldots, T\}$,

$$x_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}}\, \underbrace{\frac{x_{\tau_i} - \sqrt{1-\bar{\alpha}_{\tau_i}}\, \hat{\epsilon}_{\tau_i}}{\sqrt{\bar{\alpha}_{\tau_i}}}}_{\hat{x}_0} + \sqrt{1-\bar{\alpha}_{\tau_{i-1}} - \sigma_{\tau_i}^2}\, \hat{\epsilon}_{\tau_i} + \sigma_{\tau_i} z, \qquad z \sim \mathcal{N}(0, I), \tag{10}$$

with $\sigma_{\tau_i} = 0$ in the fully deterministic case.

4.2 Training objective I: simple loss

The *simple* objective is the standard masked denoising regression of Eq. (11) applied to the network's clean-sequence prediction:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\, x_0,\, \epsilon}\Big[\Big\| M \odot \Big(f_\theta(x_t, t, c) - x_0\Big)\Big\|_2^2 \Big/ \sum M\Big]. \tag{11}$$

This is the direct analogue of the original DDPM objective and embeds no financial domain knowledge.

4.3 Training objective II: composite finance-aware loss

Financial returns exhibit stylized facts—volatility clustering, heavy tails, skewness, and characteristic spectra—that pure mean-squared error does not explicitly reward (Cont, 2001). The *composite* objective augments Eq. (11) with auxiliary regularizers computed between the predicted clean sequence $\hat{x}_0$ and the target x_0, all masked as in Eq. (11):

$$\begin{aligned}
L_{\text{jump}} &= \Big\| (\hat{x}_{0,i+1} - \hat{x}_{0,i}) - (x_{0,i+1} - x_{0,i}) \Big\|_1 && \text{(relative jumps)}, && (12)\\
L_{\text{vol}} &= \text{SmoothL1}\Big(\text{Std}(\hat{x}_{0,\text{win}}),\ \text{Std}(x_{0,\text{win}})\Big) && \text{(volatility clustering)}, && (13)\\
L_{\text{gvol}} &= \Big|\sigma_{\text{pred}} - \sigma_{\text{true}}\Big| && \text{(global volatility)}, && (14)\\
L_{\text{kurt}} &= \Big(K_{\text{pred}} - K_{\text{true}}\Big)^2, \quad K = \mathbb{E}[z^4] - 3 && \text{(heavy tails)}, && (15)\\
L_{\text{skew}} &= \Big(\mathbb{E}[z^3_{\text{pred}}] - \mathbb{E}[z^3_{\text{true}}]\Big)^2 && \text{(skewness)}, && (16)\\
L_{\text{drift}} &= \Big(\textstyle\sum_i \Delta\hat{x}_{0,i} - \sum_i \Delta x_{0,i}\Big)^2 && \text{(terminal drift)}, && (17)\\
L_{\text{quant}} &= \textstyle\sum_{\tau \in \{0.01, 0.99\}} L_\tau\Big(q_\tau(x_0),\ q_\tau(\hat{x}_0)\Big) && \text{(extreme quantiles)}, && (18)\\
L_{\text{spec}} &= \text{SmoothL1}\Big(\hat{P}_{\text{norm}},\ P_{\text{norm}}\Big) && \text{(power spectrum)}, && (19)
\end{aligned}$$

where z denotes the standardized sequence, L_τ is the pinball loss of Wang et al. (2019) at quantile τ, and $P_{\text{norm}}[k] = P[k]/\max_k P[k]$ is the normalized discrete Fourier magnitude spectrum. The extreme-quantile pair $\tau \in \{0.01, 0.99\}$ penalizes underestimation of both tails asymmetrically, sharpening the delineation of downside risk and upside potential.

Because the auxiliary terms live on heterogeneous scales, each is normalized by an exponential moving average (EMA) of its own magnitude, and all auxiliary weights are annealed linearly from zero over a warm-up phase ($s = \min(1, \text{step}/\text{warmup})$):

$$\mathcal{L}_{\text{complex}} = \mathcal{L}_{\text{simple}} + s \sum_{k \in \mathcal{K}} \lambda_k \frac{L_k}{\text{EMA}[L_k] + \varepsilon}, \tag{20}$$

with base weights $(\lambda_{\text{jump}}, \lambda_{\text{vol}}, \lambda_{\text{gvol}}, \lambda_{\text{kurt}}, \lambda_{\text{skew}}, \lambda_{\text{drift}}, \lambda_{\text{quant}}, \lambda_{\text{spec}}) = (0.5, 3.5, 8, 5, 0.5, 2, 3, 2)$.

4.4 Heavy-tailed diffusion: the DLPM

Gaussian diffusion imposes light-tailed increments on the forward chain. The original Denoising Lévy Probabilistic Model of Shariatian et al. (2024) is isotropic. Our deployed financial-path variant instead uses coordinate-wise symmetric α-stable innovations: for return coordinate j, $\varepsilon_{t,j}^{(\alpha)} \sim \mathcal{S}\alpha\mathcal{S}(1)$, independently across j, with stability index $\alpha \in (1, 2]$. This avoids a single path-level scale draw imposing common shocks across every ordered return coordinate. The forward chain is applied coordinate-wise,

$$x_{t,j} = \gamma_t x_{t-1,j} + \sigma_t \varepsilon_{t,j}^{(\alpha)}, \qquad t = 1, \dots, T, \tag{21}$$

or, equivalently in vector notation,

$$x_t = \gamma_t \, x_{t-1} + \sigma_t \, \varepsilon_t^{(\alpha)}, \qquad t = 1, \dots, T, \tag{22}$$

under a *scale-preserving* schedule: with $\bar{\gamma}_t = \prod_{i \le t} \gamma_i$, the stability property gives the marginal

$$x_t = \bar{\gamma}_t \, x_0 + \bar{\sigma}_t \, \bar{\varepsilon}^{(\alpha)}, \qquad \bar{\sigma}_t = \left(1 - \bar{\gamma}_t^{\alpha}\right)^{1/\alpha}, \tag{23}$$

so $\bar{\gamma}_t^{\alpha} + \bar{\sigma}_t^{\alpha} = 1$ generalizes variance preservation. We derive $\gamma_t = \alpha_t^{1/\alpha}$ from the same cosine schedule as the Gaussian model, so both models traverse comparable signal-to-noise trajectories.

Computations with stable laws become Gaussian conditionally on auxiliary variables (Samorodnitsky and Taqqu, 1994). In the deployed coordinate-wise variant, each return coordinate admits the *variance-mixing* representation

$$\varepsilon_{t,j}^{(\alpha)} \overset{d}{=} \sqrt{a_{t,j}}\, Z_{t,j}, \qquad a_{t,j} \sim \mathcal{S}^{+}_{\alpha/2}, \quad Z_{t,j} \sim \mathcal{N}(0,1), \tag{24}$$

with $\mathcal{S}^{+}_{\alpha/2}$ the totally skewed positive $(\alpha/2)$-stable law. The implementation uses the Chambers–Mallows–Stuck convention equivalent to stability $\gamma = \alpha/2$, skewness $\beta = 1$, location zero, and scale $c_A = 2\cos(\pi\alpha/4)^{2/\alpha}$. For numerical stability the deployed sampler replaces $a_{t,j}$ by $\min(a_{t,j}, 100)$. Consequently, the implemented forward law is a capped stable-mixture approximation rather than an unbounded exact α-stable law; all reported model and sampler results use this same convention. Conditionally on the coordinate-wise chain $A = (a_{t,j})_{t,j}$, the forward process is Gaussian with stochastic diagonal variances. The following posterior recursion applies element-wise, with $\Sigma_{t,j}$ and $\Gamma_{t,j}$ replacing the scalar notation:

$$\Sigma_{t,j} = \sigma_t^2\, a_{t,j} + \gamma_t^2\, \Sigma_{t-1,j}, \quad \Gamma_{t,j} = 1 - \frac{\gamma_t^2\, \Sigma_{t-1,j}}{\Sigma_{t,j}}, \quad \tilde{m}_{t-1,j} = \frac{x_{t,j} - \bar{\sigma}_t\, \Gamma_{t,j}\, \varepsilon_{t,j}}{\gamma_t}, \quad \tilde{\Sigma}_{t-1,j} = \Gamma_{t,j}\, \Sigma_{t-1,j}. \tag{25}$$

Training objective III: standard DLPM loss. The network predicts the chain noise $\varepsilon_t = (x_t - \bar{\gamma}_t x_0)/\bar{\sigma}_t$. A training sample is constructed via Eq. (24),

$$a_t \sim \mathcal{S}^{+}_{\alpha/2}, \quad z \sim \mathcal{N}(0, I), \quad x_t = \bar{\gamma}_t\, x_0 + \bar{\sigma}_t \sqrt{a_t}\, z, \qquad \varepsilon_t = \sqrt{a_t}\, z, \tag{26}$$

and the *standard loss* is the L_p objective

$$\mathcal{L}_{\mathrm{DLPM}} = \mathbb{E}_{t,\,x_0,\,a_t,\,z}\Big[\Big\| M \odot \Big(\hat{\varepsilon}_\theta(x_t,t,c) - \varepsilon_t\Big)\Big\|_p\Big]. \tag{27}$$

Because α-stable noise has infinite variance for $\alpha < 2$, the exponent must respect a moment constraint; we use the *unsquared* L_2 norm—an order-one moment of $\|\varepsilon\|_2 = \sqrt{a}\,\|Z\|_2$—which is finite for all $\alpha > 1$. Shariatian et al. (2024) additionally propose a median-of-means estimator over multiple draws of a_t; the single-draw estimator proved sufficient at our scale. For numerical stability, extreme draws of a_t are clamped at $a_{\max} = 100$. In the deployed configuration, the coordinate-wise innovation supplied to the denoising chain is also symmetrically clipped at $|\varepsilon| \leq 2$, matching the $2\tanh(\cdot)$ output range of the U-Net. Thus the implemented process is a numerically truncated heavy-tailed DLPM rather than an unbounded stable law; the stable scale mixture remains the tail-generating mechanism within this explicit computational support. The controlled standard-DLPM ablation does not use the financial regularizers of Section 4.3. The final full-sample model used for the reported path and P–Q experiments is a history-aware conditional DLPM that adds a small, clipped finance-aware auxiliary term to Eq. (27). Its warm-up and scale are reported in Appendix B; this separates the theoretical DLPM objective from the engineering stabilization used in the final production run.

Sampling. *Ancestral* sampling draws the auxiliary chain A and initial noise $x_T = \bar{\sigma}_T\varepsilon^{(\alpha)}$, then iterates the Gaussian posterior of Eq. (25) with $\varepsilon_t \to \hat{\varepsilon}_\theta(x_t,t,c)$. Because the variances Σ_t are driven by heavy-tailed a_t, individual reverse trajectories can take occasional large steps—precisely the mechanism that reproduces jumps. *DLIM*, the deterministic skip-step analogue of DDIM, transports the state exactly between noise levels of a subsequence $\{\tau_i\}$:

$$x_{\tau_{i-1}} = \frac{\bar{\gamma}_{\tau_{i-1}}}{\bar{\gamma}_{\tau_i}}\Big(x_{\tau_i} - \bar{\sigma}_{\tau_i}\,\hat{\varepsilon}_{\tau_i}\Big) + \bar{\sigma}_{\tau_{i-1}}\,\hat{\varepsilon}_{\tau_i}. \tag{28}$$

5 Model Selection and Held-Out Path Evaluation

5.1 What the comparison is designed to identify

The final-split comparison holds the same chronological test set fixed across contract-only DLPM, history-aware Gaussian DDPM with simple loss, history-aware Gaussian DDPM with finance-aware loss, the frozen history-aware DLPM, historical block bootstrap, historical-drift GBM, conditional neural SDE, and Chronos-T5 zero-shot. The neural families use repeated seeds where applicable. This design separates the value of historical conditioning and the heavy-tailed diffusion mechanism from the value of a generic physical-measure baseline; downstream payoff results do not participate in model selection.

5.2 Production training and checkpoint selection

The final model uses $T = 1000$ diffusion steps, stability index $\alpha = 1.9$, a U-Net base width of 64 with channel multipliers $(1, 2, 4, 8)$, and a 128-dimensional fused condition embedding. Training uses Adam with learning rate 10^{-5}, batch size 32, cosine decay, EMA decay 0.995, and regime-balanced sample weights capped at 1.5. The cap is applied after normalization so that the realized maximum weight cannot exceed the configured ceiling. The finance-aware term is warmed up for 1,000 steps, scaled by 0.07, and clipped at 2.5 before being added to the DLPM denoising objective; model-space returns and sampled x_0 are clipped at 1.25 and 1.35, respectively. Appendix Table 16 reports every sub-weight.

Regime balancing is market-agnostic. Every training window receives a weight from the same transformation of volatility, trend, and drawdown state; country is used for stratification, not for a country-specific rule. The objective is to prevent the large mass of quiet windows from overwhelming high-volatility and reversal states while preserving the empirical market composition.

EMA-12000 achieves the best validation composite score among the pre-specified production checkpoints. It is frozen before generation of the final test archive. Selection uses path-distribution diagnostics only; no payoff result or test-set comparison is used to choose the reported checkpoint.

5.3 Full held-out path results

Table 2 reports the final-split path audit. The frozen EMA-12000 DLIM-50 model has terminal CRPS 0.0485 and path energy score 0.3368. Its aggregate signed terminal-drift error is -0.27 percentage points and its generated-to-realized volatility ratio is 0.882. As an auxiliary pooled terminal-distribution diagnostic, it obtains KS statistic 0.1004, Wasserstein distance 0.0228, and quantile–quantile R^2 of 0.8803. These statistics summarize marginal terminal-shape agreement and complement, rather than replace, the conditional ensemble scores and interval coverage in Table 2. These results accompany material improvements over the contract-only and Gaussian controls, and are consistent with contributions from both the history-aware architecture and the heavy-tailed mechanism beyond a contract-only diffusion specification; the fully budget- and loss-matched isolation of the mechanism is the $\alpha = 2$ ablation noted in Section 7. Detailed centre-dispersion diagnostics are reported separately to guide future conditional calibration.

Table 2: Final-split held-out path quality and representative controls

Model	Terminal CRPS	Energy score	90% coverage	Volatility ratio	Tail error
History-aware DLPM	0.0485	0.3368	0.704	0.882	0.0075
Contract-only DLPM	0.0548	0.3626	0.735	0.767	0.0091
Gaussian DDPM, simple loss	0.0831	0.5461	0.226	0.154	0.0124
Gaussian DDPM, finance-aware	0.1521	0.9153	0.234	0.762	0.0100
Historical block bootstrap	0.0486	0.3297	0.864	1.139	0.0066
Historical-drift GBM	0.0498	0.3397	0.897	1.246	0.0102
Conditional neural SDE	0.0579	0.3774	0.794	1.537	0.0105
Chronos-T5 zero-shot	0.0542	0.3668	0.766	1.060	0.0093

All entries use the same 6,824 chronological test windows. Neural families are seed means when more than one seed is available. Tail error is the absolute terminal-tail quantile error. The history-aware DLPM materially improves on contract-only and Gaussian controls. Historical resampling is retained as a deliberately strong unconditional control; its pooled-marginal strength motivates the state-conditional comparison reported below.

5.4 Effective tail behaviour of generated paths

The stable-law mechanism is numerically capped in the deployed model, so its economic relevance must be assessed from generated returns rather than from the unbounded theoretical α-stable law alone. Table 3 uses the first future log return at each of the 6,824 forecast origins. This forecast-aligned construction counts each realized origin once and avoids artificially multiplying observations through overlapping multi-day target windows. Generated models contribute all paths at that first step: 40 for the frozen DLPM, while each control uses its frozen evaluation ensemble.

Table 3: Effective one-step tail behaviour under the deployed numerical protocol

Source	Hill $\hat{\alpha}_{L/R}$ (5%)	Hill $\hat{\alpha}_{L/R}$ (1%)	Excess kurtosis	Tail/IQR L/R	Realized q_{01}/q_{99} exceedance (%)
Realized	2.71 / 2.20	2.93 / 2.92	9.75	2.93 / 3.33	1.01 / 1.01
History-aware DLPM	2.37 / 2.40	2.35 / 2.26	14.95	2.91 / 2.61	0.62 / 0.65
Gaussian DDPM	2.61 / 3.53	1.45 / 5.23	39.36	2.99 / 8.03	0.01 / 0.00
Historical block bootstrap	2.02 / 2.80	2.74 / 3.76	10.95	4.42 / 3.46	1.75 / 0.64
Historical-drift GBM	4.28 / 4.26	6.77 / 6.74	0.33	1.84 / 1.84	0.63 / 0.25

Hill estimates use tail magnitudes around each source median; lower values indicate slower empirical tail decay. Tail/IQR reports $(q_{50} - q_{01})$/IQR and $(q_{99} - q_{50})$/IQR. The final column applies the realized 1% left- and right-tail thresholds to each source. These are finite-sample effective-tail diagnostics under $a_t \leq 100$ and $|\varepsilon| \leq 2$, not estimates of an unbounded stable law.

The DLPM reproduces the realized left-tail-to-IQR ratio almost exactly (2.91 versus 2.93) and retains similar Hill decay on both sides. Its absolute 1% threshold exceedance rates, 0.62% on the left and 0.65% on the right, show that the deployed distribution remains somewhat narrower than realized returns despite having a suitably heavy conditional tail shape. Historical block resampling is stronger on absolute tail frequency but overstates the left tail relative to the IQR. Historical-drift GBM remains thin-tailed. The Gaussian DDPM combines a collapsed central distribution with isolated outliers: its high kurtosis therefore does not constitute calibrated heavy-tail recovery. Reporting Hill decay, tail-to-IQR ratios, and exceedance jointly prevents any single moment from driving the conclusion.

5.5 Calibration and sharpness of the 90% path fan

Coverage alone rewards arbitrarily wide intervals, whereas width alone rewards intervals that miss the observation. We therefore report both calibration and sharpness (Gneiting et al., 2007) and combine them with the central-90% interval score

$$\mathrm{IS}_{0.1}(l, u; y) = (u - l) + 20(l - y)\mathbf{1}\{y < l\} + 20(y - u)\mathbf{1}\{y > u\}. \tag{29}$$

Widths and interval scores are normalized by the observed initial level S_0. "Terminal" uses the 5th–95th percentiles of terminal returns. "Pointwise" uses the 5th–95th percentile fan at every available trading day and then averages over dates and windows. Thus neither quantity is a simultaneous whole-path coverage probability.

Table 4: Overall 90% calibration–sharpness audit

Scope	Coverage	$\lvert$coverage − 0.90$\rvert$	Mean width/S_0	Interval score/S_0
Terminal return	0.7043	0.1957	0.1642	0.4893
Pointwise path fan	0.7277	0.1723	0.1135	0.3617

The audit reuses the frozen 40-path archive; no model retraining, checkpoint reselection, or additional path generation is involved. Lower coverage error, width, and interval score are better, but width is meaningful only jointly with coverage.

Figure 2 makes this trade-off explicit. The Dow Jones fan is both sharp and almost exactly calibrated (90.08% pointwise coverage with mean width 11.46% of S_0), and the S&P 500 reaches 86.88% coverage at comparable width. Aggregate undercoverage is concentrated in CSI 1000, CSI 500, low-trend, deep-drawdown, and low-volatility windows. Since width varies much less than coverage across indices and volatility states, a single global widening factor would over-widen already calibrated markets. The evidence instead points to state- and market-dependent conditional dispersion as the relevant calibration frontier.

Figure 4 shows one conditional fan per index. To avoid visual cherry-picking, each panel uses the window at the median rank of absolute terminal-median error for that index. The

Table 5: Pointwise 90% fan calibration by index, tenor, and pre-window state

Dimension	Group	Windows	Coverage	Mean width/S_0
Index	CSI 1000	827	0.4834	0.1129
Index	CSI 300	827	0.7304	0.1111
Index	CSI 500	827	0.5612	0.1127
Index	Dow Jones	879	0.9008	0.1146
Index	NASDAQ	879	0.7502	0.1153
Index	Russell 2000	879	0.7594	0.1159
Index	S&P 500	879	0.8688	0.1150
Index	SSE Composite	827	0.7439	0.1102
Tenor	30 days	2,796	0.7507	0.0756
Tenor	90 days	2,316	0.7025	0.1188
Tenor	180 days	1,596	0.7154	0.1635
Tenor	365 days	116	0.8457	0.2339
Trend	Low	2,270	0.6394	0.1131
Trend	Middle	2,280	0.7694	0.1130
Trend	High	2,274	0.7740	0.1145
Drawdown	Deep	2,276	0.6450	0.1117
Drawdown	Middle	2,274	0.7671	0.1133
Drawdown	Shallow	2,274	0.7710	0.1155
Volatility	Low	2,272	0.6458	0.1178
Volatility	Middle	2,277	0.7288	0.1077
Volatility	High	2,275	0.8083	0.1151

States are formed from information available before the forecast window, using within-index terciles of 60-day trend, current drawdown, and pre-window realized volatility. The machine-readable supplement also reports terminal coverage, coverage error, and interval scores for every group.

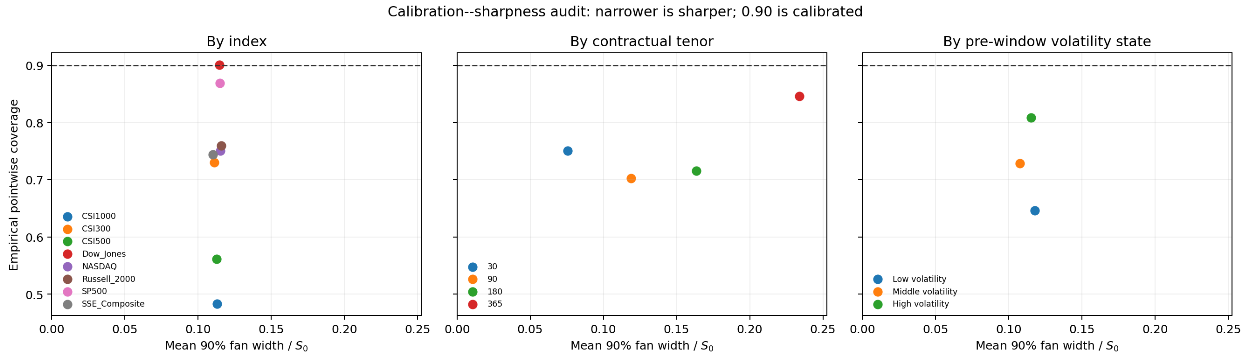

Figure 2: Calibration–sharpness diagram for the central 90% pointwise path fan. Moving left indicates sharper intervals; moving toward the dashed 0.90 line indicates better calibration. Wide intervals with high coverage are not automatically preferable, because the interval score penalizes excess width.

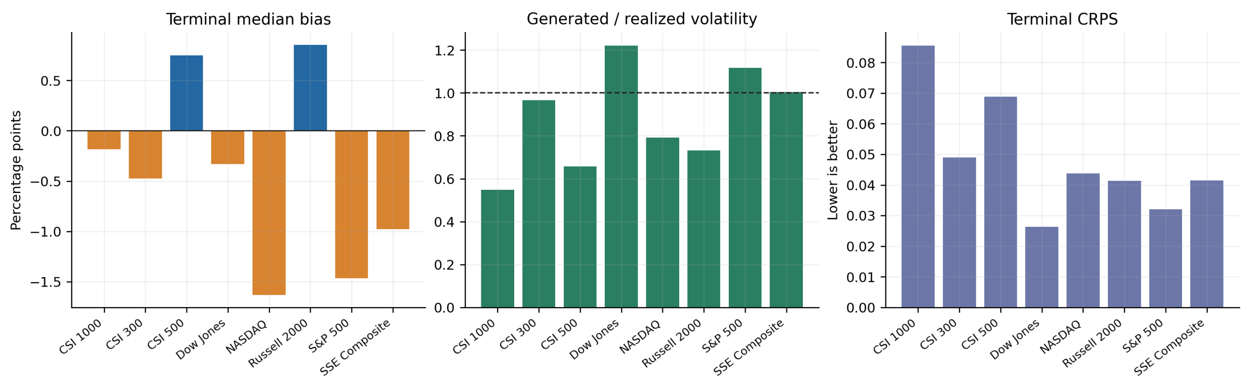

Figure 3: Cross-index held-out path quality. The panels separate quantities with different units and directions: terminal bias, volatility ratio, and terminal CRPS are not placed on one common axis.

realized path begins at the same observed S_0 as every generated path. The fans expand with horizon, change scale across markets, and often place the realized trajectory close to the conditional median. Sharp ex-post jumps can leave the central interval, as they should in a probabilistic forecast; the relevant question is whether the distribution assigns sensible mass around such events across the population, which is measured in Table 2.

5.6 Interpreting center accuracy and uncertainty

The audit separates aggregate level accuracy from conditional calibration. Paths are exactly anchored at observed S_0, the signed drift error is small, and the DLPM materially improves on the Gaussian controls. The interval diagnostics provide a clear calibration target for deployment-specific post-processing. The downstream P–Q experiment uses all generated paths as a payoff-sensitivity diagnostic, complementing rather than replacing an option-surface-calibrated workflow.

5.7 Accelerated sampling

The production sampler uses 50 DLIM steps rather than the complete 1,000-step reverse chain. This makes the full 6,824-window, 40-path archive and repeated payoff evaluation computationally feasible. Controlled sampler audits show that acceleration changes higher moments more than the conditional center. Accordingly, all headline statistical and economic numbers use the same DLIM-50 archive; full-chain results are mechanism diagnostics, not silently mixed into the game. This common-path discipline is more important than choosing a sampler separately for every favorable metric.

6 The Trader–Market-Maker P–Q Quoting Game

6.1 An independently constructed risk-neutral benchmark

For each index, the Q benchmark begins with a Student-t GARCH(1,1) volatility recursion (Bollerslev, 1986),

$$h_{t+1} = \omega + \alpha \epsilon_t^2 + \beta h_t, \tag{30}$$

$$z_{t+1} \sim t_\nu, \qquad \epsilon_{t+1} = \sqrt{h_{t+1}} z_{t+1}. \tag{31}$$

Parameters are fitted to a deduplicated chronological return series through the validation period. GARCH persistence and Student-t tails make this a materially stronger opponent than constant-volatility GBM.

Historical GARCH fitting does not itself define Q. We therefore transform the innovation law and conditional drift explicitly. Let z denote a standardized Student-t draw. A symmetric raw cap is solved so that, after truncation and re-standardization,

$$\widetilde{z} = \frac{\operatorname{clip}(z, -c_\nu, c_\nu)}{\sqrt{\operatorname{Var}[\operatorname{clip}(z, -c_\nu, c_\nu)]}}, \qquad |\widetilde{z}| \le 8, \qquad \operatorname{Var}(\widetilde{z}) = 1. \tag{32}$$

This step is important: without post-truncation normalization, the simulated innovation variance could be far below the fitted GARCH conditional variance for low effective degrees of freedom.

The GARCH model is fitted to daily log returns (temporarily expressed in percentage points for numerical estimation and converted back afterward). Accordingly, h_{t+1} is a one-trading-day conditional log-return variance, not an annualized variance. For annualized risk-free rate r_t, annualized dividend or carry yield q_t, and $\Delta = 1/252$, define

$$\kappa_t = \log \mathbb{E}\left[\exp\left(\sqrt{h_{t+1}}\, \widetilde{z}_{t+1}\right) \mid \mathcal{F}_t\right]. \tag{33}$$

The RN-Q level process is

$$S_{t+1} = S_t \exp\left((r_t - q_t)\Delta - \kappa_t + \sqrt{h_{t+1}}\, \widetilde{z}_{t+1}\right). \tag{34}$$

By construction,

$$\mathbb{E}^Q[e^{-(r_t - q_t)\Delta} S_{t+1} \mid \mathcal{F}_t] = S_t. \tag{35}$$

The log-mgf is evaluated numerically under the same capped, unit-variance innovation law used for simulation. Empirical audits across all indices and 20-, 60-, 120-, and 240-day horizons produce mean discounted-price ratios close to one.

The archived experiment sets $q_t = 0$. All eight underlyings are observed as price-index level series, but the conditioning data do not contain a matched dividend-yield or futures-basis series. RN-Q is therefore an internally risk-neutral, zero-carry benchmark under its specified law, rather than a market-calibrated index-option measure. The code exposes q_t explicitly for future carry-aware experiments. A market-implied Q would additionally require dividend or futures data and option-surface calibration. This qualification does not alter the present martingale audit, which tests the stated $q_t = 0$ benchmark.

6.2 Why the comparison is informative

P and Q encode different information. The DLPM learns a flexible, history-conditioned physical distribution with heavy-tailed diffusion noise. RN-Q imposes a discounted martingale while carrying GARCH volatility persistence and Student-t innovations. Comparing their payoff expectations therefore asks whether information learned under P survives a disciplined, nontrivial quoting benchmark.

For payoff $H(S_{0:T})$, define

$$V_P = e^{-rT}\mathbb{E}_P[H(S_{0:T})], \tag{36}$$

$$V_Q = e^{-rT}\mathbb{E}_Q[H(S_{0:T})]. \tag{37}$$

Price-level products use the symmetric total spread parameter g,

$$V_Q^{ask} = V_Q(1 + g/2), \qquad V_Q^{bid} = V_Q(1 - g/2). \tag{38}$$

Thus $g = 0.10$ denotes a total 10% bid–ask band, or approximately 5% on each side; it is not a 10% transaction cost and not a guaranteed Q profit rate. Structured rate-style products map the same stress parameter into their documented notional-rate spread conventions.

The materiality threshold is stated in economically comparable spot units: $\tau_S = 0.01$. The P side buys when $V_P - V_Q^{ask} > \tau_S S_0$ and sells when $V_Q^{bid} - V_P > \tau_S S_0$. If the realized discounted payoff is V_{real}, then

$$\text{P\&L} = \begin{cases} V_{\text{real}} - V_Q^{ask}, & \text{P buys,} \\ V_Q^{bid} - V_{\text{real}}, & \text{P sells,} \\ 0, & \text{no trade.} \end{cases} \tag{39}$$

The $1\% S_0$ threshold filters economically immaterial valuation gaps and is separate from g. There is no additional hidden transaction-cost deduction in the current experiment.

Because quotes are Monte Carlo estimates, we also report a sampling-aware execution rule. For side-specific discounted payoff samples, let $\widehat{\text{SE}}_P$ and $\widehat{\text{SE}}_Q$ denote the sample standard deviation divided by the square root of the number of paths. A quote passes the conservative filter only if

$$|V_P - V_Q^{\text{side}}| > \tau_S S_0 + k\sqrt{\widehat{\text{SE}}_P^2 + \widehat{\text{SE}}_Q^2}. \tag{40}$$

The primary table retains the economically transparent $k = 0$ rule; $k = 1$ and $k = 1.96$ are pre-specified robustness filters against threshold crossing caused by quote noise.

6.3 Final payoff protocol and alignment audit

The final economic audit uses vanilla, arithmetic-average Asian, and floating-strike lookback calls. These standard payoff families provide a progression from terminal to average and running-extremum dependence while remaining naturally comparable after normalization by initial spot. For each held-out contract window, the frozen history-aware DLPM supplies 40 P paths and the independently calibrated RN-Q benchmark supplies 256 paths. P and Q arrays are joined by asset and chronological window identifier; file-level checks verify S_0, rate, tenor, and realized future path before payoff evaluation. Results include every window: a non-executed quote contributes zero P&L. This unconditional convention avoids the selection effect that occurs when results are averaged only over executed trades.

For reproducibility, let a window contain the observed initial level and all available subsequent trading-day observations, $S_{0:m} = (S_0, S_1, \ldots, S_m)$, where m is the stored `actual_trading_days`; no unavailable date is padded. All three standard contracts are at the money, $K = S_0$, and use

$$H_{\text{Vanilla}}(S_{0:m}) = (S_m - K)^+, \tag{41}$$

$$H_{\text{Asian}}(S_{0:m}) = \left(\frac{1}{m+1}\sum_{i=0}^{m} S_i - K\right)^+, \tag{42}$$

$$H_{\text{Lookback}}(S_{0:m}) = \left(S_m - \min_{0\le i\le m} S_i\right)^+. \tag{43}$$

The quoted value is $e^{-rm/252}$ times the Monte Carlo mean payoff, using the annualized window rate r. The initial RN-Q Delta diagnostic uses a central spot bump $\epsilon = 0.001$,

$$\Delta_0 = \frac{V_Q(S_0(1+\epsilon)) - V_Q(S_0(1-\epsilon))}{2\epsilon S_0}, \tag{44}$$

with common random numbers: the same RN-Q innovations are rescaled under both spot bumps. The resulting index position is opened once and held to maturity; it is not a dynamic replication strategy.

6.4 Structured-product stress-test scenarios

The scope of the paper extends beyond smooth options. We retain two explicitly specified structured-product scenarios as stress tests for future P–Q evaluation: an accumulator and a snowball/autocall note. Their purpose is not to produce a single number comparable with a vanilla option price. Rather, they expose the generated path distribution to repeated observations, early termination, leverage after a drawdown, and clustered tail events—the features for which a terminal marginal is least informative.

The mean quote standard errors, normalized by S_0, are 0.66%, 0.38%, and 0.63% on the P side and 0.39%, 0.20%, and 0.41% on the Q side for Vanilla, Asian, and Lookback, respectively. Table 7 therefore subjects the result to the explicit measurement-error filter

Table 6: Structured-product stress-test specifications

Scenario	Frozen payoff convention
Accumulator	Daily observations; purchase strike $0.85S_0$; knock-out at $1.05S_0$; two-times quantity below strike; scheduled payoff normalized by the number of remaining observations.
Snowball/autocall	Five-day knock-out observations at $1.05S_0$; knock-in barrier $0.80S_0$; annual coupon rate 15%; maturity loss conditional on a knock-in without knock-out.

These are scenario specifications, not smooth-option price-equivalent contracts. A valid empirical report must separately state coupon and loss distributions, knock-in and knock-out frequencies, expected shortfall after knock-in, and Monte Carlo concentration. They are therefore not pooled with the normalized Vanilla/Asian P&L headline table.

in Equation (40). The stricter rules sharply reduce execution, as they should, while the unconditional Vanilla and Lookback diagnostics remain positive.

Table 7: Monte Carlo quote-uncertainty audit at $g = 0.05$ and $\tau_S = 1\%S_0$

Filter	Contract	Trade rate	Unconditional P&L/S_0
$k = 0$	Vanilla / Asian / Lookback	30.95% / 13.69% / 35.04%	0.91% / 0.36% / 0.28%
$k = 1$	Vanilla / Asian / Lookback	11.24% / 4.35% / 14.63%	0.44% / 0.12% / 0.17%
$k = 1.96$	Vanilla / Asian / Lookback	4.05% / 1.64% / 5.95%	0.25% / 0.04% / 0.08%

The filter adds k joint Monte Carlo standard errors to the fixed materiality threshold. P uses 40 paths and RN-Q uses 256 paths in every row.

At the stringent $k = 1.96$ setting, varying τ_S from 0.5% to 1.0% and 1.5% changes the Vanilla trade rate from 7.44% to 4.05% and 2.51%, with unconditional P&L of 0.34%, 0.25%, and 0.19%. The corresponding Lookback figures are 11.23%, 5.95%, and 2.95% for execution and 0.11%, 0.08%, and 0.04% for P&L. Asian is more sensitive, falling from 0.10% to 0.04% and approximately zero. Thus the smooth-option evidence is not based on an unreported 40% threshold, and its strongest residual is concentrated in Vanilla and the running-extremum contract.

The present aligned economic archive reports the three standard contracts below. The one-time delta check is reported only for Vanilla and Asian because a static delta is not a complete hedge for a running-extremum payoff. The structured scenarios are evaluated separately with the same unique-key P/Q alignment. Table 8 reports normalized-notional scenario statistics, not option prices. The P generator produces higher mean scenario payoff than RN-Q for both products while retaining realistic early-termination behavior: the realized accumulator mean lies below both generated means, whereas the realized snowball mean lies between the P and Q means; realized knock-out and coupon frequencies are of the same order as both generated mechanisms.

Table 8: Aligned structured-product scenario audit on 6,824 windows

Scenario	P mean	RN-Q mean	Realized mean	P–Q gap	P/Q/realized KO rate	P/Q/realized coupon rate
Accumulator	11.01%	9.80%	9.33%	1.21pp	45.82% / 43.50% / 52.71%	–
Snowball	2.45%	1.86%	1.94%	0.59pp	42.40% / 39.09% / 49.68%	99.96% / 97.52% / 98.49%

Each figure is a normalized-notional scenario payoff or event frequency. The reported P archive has 40 paths per window and RN-Q has 256. Results are not combined with the smooth-option P&L table because the contracts have different economic units and loss mechanisms.

The positive mean and event-rate evidence makes these products useful stress-test cases for a path generator, rather than merely background examples. At the same time, the path-level tail audit finds that the current P generator understates rare Snowball knock-ins relative to the realized panel. Accordingly, the evidence supports structured-product scenario generation and average-payoff diagnostics, but not yet standalone tail-risk pricing for knock-in products.

The main dealer-spread point is $g = 0.05$, with $g = 0.10$ and $g = 0.20$ reported as pre-specified stress cases. All reported smooth-option diagnostics use the same complete-window archive with 40 P paths per window.

6.5 Final P–Q payoff results

Table 9 reports the final normalized results. The P and Q archives are joined with the unique key (asset, start date, S_0, tenor, rate), rather than by row position; the post-join initial-price check is exact to numerical tolerance. At $g = 0.05$ and $\tau_S = 1\% S_0$, unconditional Vanilla P&L is 0.91% of S_0 and Asian P&L is 0.36%. The associated execution rates are 30.95%

and 13.69%, respectively. Buy and sell directions are both represented, but the positive raw gap is predominantly a buy-side contribution. This motivates the drift-neutral and hedged counterfactuals below.

Table 9: Final P–Q payoff diagnostic at $g = 0.05$ and $\tau_S = 1\% S_0$

Contract	P&L/S_0	Trade rate	Win rate	P-buy share	Buy P&L/S_0	P-sell share	Sell P&L/S_0
Vanilla call	0.91%	30.95%	61.27%	63.35%	0.86%	36.65%	0.06%
Asian call	0.36%	13.69%	59.74%	78.16%	0.33%	21.84%	0.03%
Floating-strike lookback call	0.28%	35.04%	55.96%	33.88%	0.61%	66.12%	-0.33%

P&L is unconditional across all 6,824 windows and normalized by initial spot. A no-trade window contributes zero. Buy/sell shares condition on an executed trade. Buy and sell P&L columns are unconditional contributions: trade rate × direction share among active trades × mean P&L of that direction; their sum equals total P&L/S_0.

The Lookback result is especially useful as a simple exotic diagnostic: it depends on the running minimum rather than only a terminal or average level. Its 66.12% sell-side share is materially higher than for Vanilla and Asian, so the positive unconditional value is not merely a long-trend signal. Because the payoff depends on extrema, its sampling uncertainty is wider; it is therefore reported alongside, rather than folded into, the static-delta table.

Table 10: One-time RN-Q Delta-hedged payoff diagnostic at $g = 0.05$ and $\tau_S = 1\% S_0$

Contract	Trade rate	Unhedged P&L/S_0	Hedged P&L/S_0	Hedged active win rate
Vanilla call	30.95%	0.91%	0.33%	60.84%
Asian call	13.69%	0.36%	-0.04%	49.57%

An RN-Q central-difference Delta is set at inception and held to maturity. This is a directional-exposure diagnostic, not a full dynamic hedge or a transaction-cost-inclusive trading strategy.

6.6 Drift-neutral counterfactual and economic interpretation

The P–Q diagnostic intentionally compares a learned physical distribution with an independently constructed martingale benchmark. Its payoff gap therefore contains several economically meaningful channels: physical risk-premium exposure, state-dependent volatility

Table 11: Dealer-spread sensitivity of unconditional normalized P&L

	$g = 0.05$		$g = 0.10$		$g = 0.20$	
Contract	P&L/S_0	Trade	P&L/S_0	Trade	P&L/S_0	Trade
Vanilla	0.91%	30.95%	0.83%	27.49%	0.66%	21.61%
Asian	0.36%	13.69%	0.33%	12.00%	0.25%	9.06%
Floating-strike lookback	0.28%	35.04%	0.21%	29.94%	0.17%	20.97%

The parameter g is the total symmetric dealer spread in Equation (38), not a transaction-cost estimate.

and tail-shape differences, and conditional path geometry. We decompose this gap with two conservative counterfactuals. Terminal-drift neutralization moves Vanilla and Asian P&L from 0.91% and 0.36% to −0.95% and −0.32%; a one-time initial RN-Q delta hedge yields 0.33% and −0.04%. Thus physical direction exposure is an important component of the raw P–Q gap. At the same time, the positive static-hedged Vanilla residual shows that the selected conditional scenarios retain a non-directional payoff component under this diagnostic. The Asian residual is close to zero. This evidence is not used to select the model: the P–Q module is a transparent payoff-sensitive stress test, with no option price or downstream payoff entering DLPM training or checkpoint selection.

6.7 Dependence-aware uncertainty

We reconstruct P&L for every asset–date–contract cell and jointly resample common calendar blocks across all eight indices. This preserves cross-market shocks and the serial overlap created by rolling windows. Table 12 reports the $g = 0.05$, $\tau_S = 1\% S_0$ intervals. The 20- and 60-day blocks are primary; 90–150 days are medium-horizon checks. The 180- and 252-day entries use only three effective blocks and are presented as long-dependence sensitivities.

Table 12: Joint calendar-block bootstrap, unconditional P&L/S_0 at $g = 0.05$ and $\tau_S = 1\%S_0$

Block days	Effective blocks	Vanilla 95% CI	Asian 95% CI	Lookback 95% CI	Role
20	27	[0.32%, 1.55%]	[0.08%, 0.73%]	[-0.02%, 0.70%]	Primary
60	9	[0.10%, 1.73%]	[-0.03%, 0.79%]	[-0.06%, 0.71%]	Primary
90	6	[0.01%, 2.18%]	[-0.04%, 0.97%]	[-0.14%, 0.81%]	Medium horizon
120	5	[-0.02%, 1.66%]	[0.03%, 0.70%]	[-0.16%, 0.76%]	Medium horizon
150	4	[0.00%, 2.08%]	[0.03%, 0.84%]	[-0.29%, 1.11%]	Medium horizon
180	3	[0.11%, 2.08%]	[0.01%, 0.95%]	[-0.01%, 0.90%]	Long sensitivity
252	3	[0.07%, 1.04%]	[-0.01%, 0.40%]	[0.33%, 0.41%]	Long sensitivity

Each interval uses 2,000 joint calendar-block replications. The long-block results should not be treated as decisive because the effective number of blocks is small. Intervals are calculated from S_0-normalized window P&L, so indices have equal economic weight rather than being weighted by their index level.

6.8 State-conditional comparison

The pooled bootstrap comparison is intentionally retained as a demanding nonparametric control: resampling reuses realized historical moves and is therefore naturally strong on unconditional marginal diagnostics. It cannot, however, distinguish a quiet start from a trend reversal or a drawdown. Since the central DLPM claim is conditional generation, Table 13 stratifies the same frozen test set using pre-window, within-index terciles. The DLPM has lower point-estimate terminal CRPS in low-trend and deep-drawdown states, the two settings in which conditional path stress testing is most consequential. Against unconditional resampling these small paired differences are descriptive: joint calendar-block intervals cross zero and Holm-adjusted p-values across the six pre-specified state slices equal one. This motivates the more direct state-matched counterfactual reported next.

Motivated by conditional predictive-ability evaluation (Giacomini and White, 2006), we test the paper's conditional claim with a separate, no-training state-matched empirical control. Its construction was specified without observing target paths: for each test window it selects the 40 nearest training windows with the same index and contractual tenor, using only standardized pre-window 60-day realized volatility, trend, and current drawdown. It never observes the test target. Table 14 shows that the DLPM lowers aggregate CRPS by 23.3%; the paired difference is negative at every joint calendar-block length from 20 through 150

Table 13: Regime-stratified terminal CRPS: frozen DLPM versus historical block bootstrap

Pre-window state (within-index tercile)	Windows	DLPM CRPS	Bootstrap CRPS
Low 60-day trend	2,270	0.06464	0.06562
Middle 60-day trend	2,280	0.04455	0.04355
High 60-day trend	2,274	0.03639	0.03671
Deep current drawdown	2,276	0.06044	0.06194
Middle current drawdown	2,274	0.04316	0.04342
Shallow current drawdown	2,274	0.04192	0.04047

Lower CRPS is better. Regimes are formed from information observable at the contract start, separately within each index. Relative DLPM improvements are 1.49% in low-trend and 2.41% in deep-drawdown windows. Historical block resampling has a mechanical advantage for linear trend continuation because it directly reuses realized trend blocks; nevertheless, the DLPM remains competitive in high-trend windows and is more informative in stress states where the pre-window volatility, drawdown, and reversal context matters. The paired state-slice differences are not statistically distinguishable after joint calendar-block resampling and Holm correction. This table compares against an unconditional replay mechanism; it does not test the direct state-matched counterfactual in Table 14.

days.

Table 14: Terminal CRPS against the state-matched empirical control

Slice	Windows	DLPM CRPS	State-matched CRPS
All windows	6,824	0.04851	0.06329
Low / middle / high trend	2,270 / 2,280 / 2,274	0.06464 / 0.04455 / 0.03639	0.08368 / 0.05244 / 0.05380
Deep / middle / shallow drawdown	2,276 / 2,274 / 2,274	0.06044 / 0.04316 / 0.04192	0.07826 / 0.05667 / 0.05491
China / United States	3,308 / 3,516	0.06156 / 0.03624	0.08279 / 0.04494

Lower is better. The overall DLPM-minus-control difference is -0.01477. The 95% joint calendar-block intervals are $[-0.01760, -0.00727]$ at 20 days, $[-0.01878, -0.00475]$ at 60 days, $[-0.02111, -0.00239]$ at 90 days, $[-0.01928, -0.00320]$ at 120 days, and $[-0.01759, -0.00344]$ at 150 days (2,000 replications).

7 Limitations and Scope

Six boundaries qualify the reported results; each is stated where it arises and consolidated here. First, the deployed sampler is numerically truncated: auxiliary draws are capped at $a_t \leq 100$ and coordinate-wise innovations at $|\varepsilon| \leq 2$, so every reported statistic describes an effective, finite-support heavy-tailed process rather than an unbounded stable law (Section 4.4,

Table 3). Second, the generator is under-dispersed at the pooled level: overall pointwise fan coverage is 0.7277 against a 0.90 target, with the shortfall concentrated in CSI 1000, CSI 500, and low-trend, deep-drawdown, low-volatility windows; because fan width varies far less than coverage across these groups, a single global widening factor would over-widen already calibrated markets, and state-dependent conditional-width calibration is the identified frontier. Third, the two Gaussian diffusion controls share the backbone and conditioning but use a separately implemented auxiliary-loss stack; since $\alpha = 2$ recovers the Gaussian chain exactly within the DLPM parameterization, a fully budget- and loss-matched α-ablation is the cleanest isolation of the heavy-tailed mechanism's marginal contribution and is left to future work. Fourth, the RN-Q benchmark is an internally consistent zero-carry construction: no option-surface, dividend, or futures-basis information enters the audit, so P–Q values measure disagreement with a stated benchmark law rather than mispricing against market quotes. Fifth, the economic results are belief-based diagnostics under pre-specified spread and materiality conventions, not transaction-cost-inclusive trading claims; joint calendar-block intervals for the Asian and Lookback contracts cross zero at several block lengths (Table 12). Sixth, the structured-product module supports scenario generation and average-payoff diagnostics but not standalone tail-risk pricing for knock-in products, whose rare-event frequencies remain understated relative to the realized panel.

8 Conclusion

This paper develops a history-aware financial adaptation of Denoising Lévy Probabilistic Models for conditional path generation and evaluates it under a broad, chronology-respecting cross-market protocol. The contribution is theoretical, computational, and empirical: it combines an α-stable diffusion mechanism with a U-Net conditioned on long return histories and financial state variables, and it evaluates one frozen generator over 6,824 out-of-time windows from eight Chinese and U.S. indices.

The final audit supports a substantive contribution. The history-aware DLPM improves terminal CRPS from 0.0548 for contract-only DLPM and 0.0831 for the simple Gaussian diffusion control to 0.0485, with path energy score 0.3368, generated-to-realized volatility ratio 0.882, and an aggregate signed terminal-median bias of -0.46pp. These gains arise from a single global model spanning eight Chinese and U.S. indices, long historical conditioning, state-aware sampling weights, and a heavy-tailed diffusion mechanism. The effective-tail audit locates the mechanism's contribution within its numerical support: generated first-step returns reproduce the realized left-tail-to-IQR ratio almost exactly (2.91 versus 2.93) with similar two-sided Hill decay, while remaining narrower than realized returns in absolute exceedance—the same under-dispersion that the calibration audit localizes to specific indices and pre-window states.

The conditional comparison sharpens the interpretation of the bootstrap result. Unconditional historical resampling is deliberately difficult to beat on pooled marginal statistics because it directly reuses realized return blocks; its overall paired CRPS difference from DLPM is statistically indistinguishable from zero. The more relevant state-matched empirical control uses the same index, tenor, volatility, trend, and drawdown information but no neural generator. Against that direct counterfactual, DLPM reduces terminal CRPS from 0.0633 to 0.0485, a 23.3% improvement, with joint calendar-block intervals excluding zero at 20, 60, 90, 120, and 150 days. This identifies the value of learning a continuous conditional path law rather than merely replaying nearby historical states.

The downstream P–Q audit complements, rather than validates, the path diagnostics. At $g = 0.05$ and $\tau_S = 1\% S_0$, raw unconditional P&L is 0.91% of spot for Vanilla, 0.36% for Asian, and 0.28% for floating-strike Lookback calls. Lookback is a particularly useful simple-exotic check because its payoff depends on a running extremum and its positive result contains a majority sell-side contribution. The one-time RN-Q Delta-hedged diagnostic retains 0.33% for Vanilla while Asian is close to zero at -0.04%. Terminal-drift-neutral counterfactuals show that physical direction remains an important component of the raw values. The aligned

Accumulator and Snowball scenarios provide a demanding test of payoff, knock-out, and knock-in mechanics; they are reported as risk scenarios rather than as static-hedge trading claims. This makes the payoff module more informative than a single unqualified backtest without conflating P–Q disagreement with economic alpha. An explicit quote-error audit further shows that adding one joint Monte Carlo standard error to the 1% materiality threshold leaves positive unconditional P&L of 0.44%, 0.12%, and 0.17% for Vanilla, Asian, and Lookback, while reducing execution to 11.24%, 4.35%, and 14.63%. The economic module thus documents payoff relevance under conservative selection without serving as the model-selection criterion.

The resulting framework is useful for conditional scenario generation, cross-market stress testing, payoff-sensitive structured-product analysis, and belief-based derivative research. Future work can extend the same engineering stack with conditional-width calibration, sampler distillation, and a direct connection between the physical generator and an option-surface-calibrated market Q.

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The authors report there are no competing interests to declare.

Data availability statement

The implementation, frozen experiment configuration, audit scripts, and the compact result data underlying all tables and figures in this article are openly available in the companion repository at https://github.com/Junchi-Shen/DLPM. Large binary artifacts, including model checkpoints and generated path archives, are archived on Hugging Face at https://huggingface.co/Islandshen/DLPM. Raw market index series were obtained from public providers (AKShare for the Chinese indices and Yahoo Finance for the U.S. indices) and remain subject to those providers' redistribution terms; the repository documents the tickers, transformations, and chronological split boundaries required to rebuild the experimental panel from the original sources.

Declaration of generative AI use

During the preparation of this work the authors used generative AI coding and editing assistants (OpenAI Codex and Anthropic Claude Code, 2025–2026 releases) to assist with software implementation, experiment tooling, and language editing. All research design, model specifications, analyses, results, and conclusions are the authors' own; the authors reviewed and verified all generated content and take full responsibility for the content of this manuscript.

A Network Architecture

Backbone. We use a one-dimensional conditional U-Net (Ronneberger et al., 2015) with base width 64, channel multipliers $(1, 2, 4, 8)$, dropout 0.1, and 17.0 million parameters. The encoder consists of residual convolutional blocks and downsampling layers that halve sequence length while increasing feature dimension; skip connections preserve encoder features for the decoder, which restores resolution by upsampling and concatenation. Linear attention operates at each resolution and full self-attention at the bottleneck. Each residual block is

$$\mathrm{ResBlock}(x) = x + \mathrm{Conv}\Big(\mathrm{SiLU}(\mathrm{Norm}(\mathrm{Conv}(x)))\Big), \tag{45}$$

with conditioning injected via feature-wise scale–shift modulation.

Time embedding. Diffusion steps enter through sinusoidal embeddings (Vaswani et al., 2017),

$$\mathrm{TimeEmb}(t) = \Big[\sin(\omega_1 t), \cos(\omega_1 t), \ldots, \sin(\omega_{d/2}\, t), \cos(\omega_{d/2}\, t)\Big], \qquad \omega_i = 10000^{-2i/d}. \quad (46)$$

Condition embedding. The two categorical identifiers pass through learned embedding tables (country: 64 dimensions; underlying index: 128 dimensions). The numerical stream contains the contract variables and history-aware features: 60/252-day returns, trend, drawdown, realized and downside volatility, volatility paths, reversal and validity indicators. It is projected with LayerNorm and SiLU; the streams are fused by learned weighted concatenation and a two-layer MLP into a 128-dimensional condition vector,

$$\mathrm{CondEmb}(c) = \mathrm{MLP}\Big([\, w_n\, \phi_n(c_{\mathrm{num}});\ w_c\, E_{\mathrm{cty}}(c_{\mathrm{country}});\ w_x\, E_{\mathrm{idx}}(c_{\mathrm{index}})\,]\Big) \in \mathbb{R}^{128}, \quad (47)$$

which modulates every residual block together with the time embedding. The index embedding is deliberately the widest stream, as underlying identity carries the strongest style information.

B Training and Sampling Hyperparameters

This appendix consolidates the frozen training, sampling, and finance-regularization choices used by every reported table. No alternative exploratory configuration is mixed into the manuscript protocol.

Table 15: Hyperparameters of the final clean-split history-aware DLPM run

Diffusion steps T	1000 (cosine schedule; final clean-split run)
Stability index α (DLPM)	1.9
DLPM loss	unsquared L_2, single-draw a_t, clamp $a_{\max} = 100$, $\lvert\varepsilon\rvert \leq 2$
Optimizer	Adam, learning rate 10^{-5}, cosine decay
Training steps / batch size	16,000 / 32; 12,000-step EMA selected on validation
EMA decay	0.995
Finance-aware auxiliary loss	clipped at 2.5, scale 0.07, warm-up 1,000 steps
Q innovation law	symmetric Student-t, unit-variance re-standardized with final cap $\lvert z\rvert \leq 8$
Q martingale correction	numerical state-dependent log-mgf
Return scale s_{vol}	0.09
Sequence length	252 (masked, variable tenor)
Sampling (path and economic audit)	50-step DLIM, 40 P paths per window, EMA-12000 weights
Sampling (accelerated)	DLIM / DDIM, 50 steps

Table 16: Finance-aware regularizer weights in the frozen configuration

Term	Weight	Term	Weight	Term	Weight
Pointwise x_0	2.50	Return range	5.00	Terminal drift	0.70
Cumulative return	0.70	Global volatility	3.00	Mean absolute return	5.00
Volatility clustering	0.50	Squared-return ACF	0.35	Return ACF(1)	0.20
Return ACF(5)	0.10	Leverage effect	0.10	Tail quantile	0.10
Absolute-tail quantile	4.50	Tail/IQR ratio	0.05	Exceedance 0.5	1.00
Exceedance 1 / 2	0.60 / 0.20	Drawdown	0.35	Path log quantile	0.50
Path absolute quantile	0.75	Path terminal quantile	0.90	Path drift	0.35

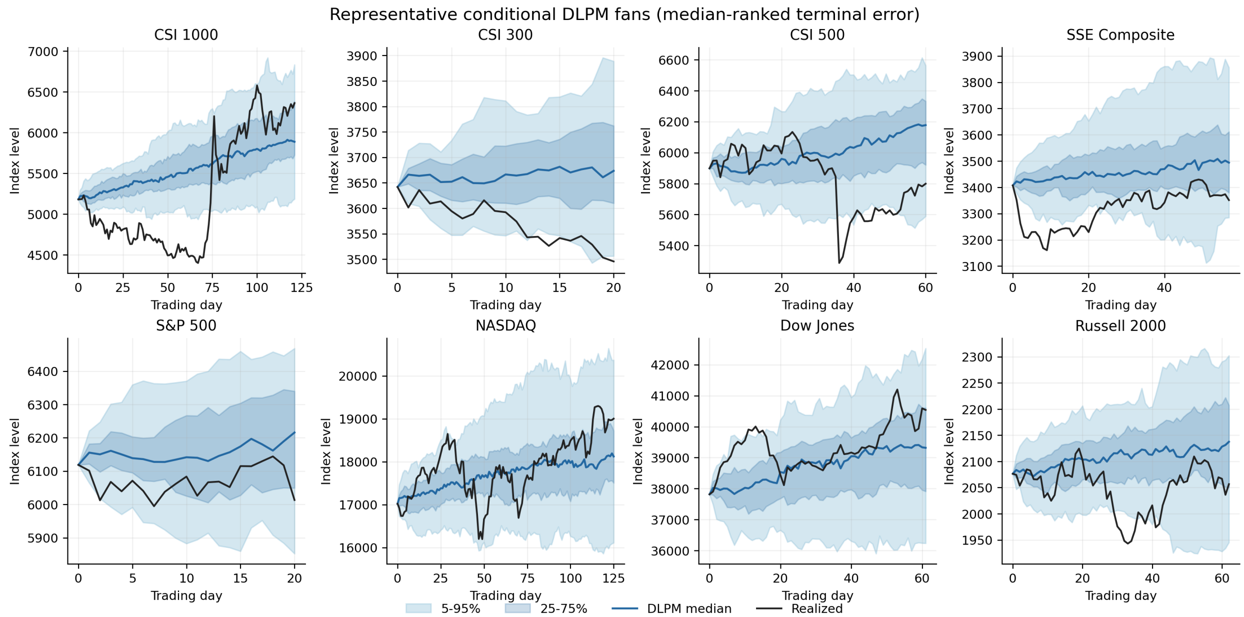

Figure 4: Representative DLPM-only conditional fans for all eight indices. Panels are selected mechanically by median-ranked terminal error. Shading shows 5–95% and 25–75% intervals; blue and black lines are the generated median and held-out realization.

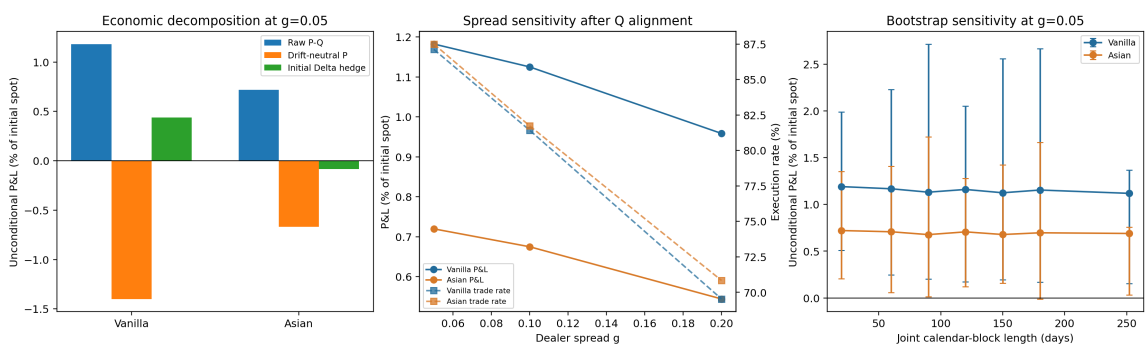

Figure 5: Aligned economic audit. The first panel separates the raw P–Q diagnostic from terminal-drift-neutral and one-time initial-delta-hedged counterfactuals. The remaining panels show dealer-spread sensitivity and dependence-aware bootstrap intervals. All displayed P&L values are unconditional and normalized by initial spot.

C Reproducibility and Data Availability

The clean companion repository contains the model, Gaussian and empirical baselines, RN-Q construction, payoff definitions, state-matched audit, calendar-block inference, frozen configuration, and scripts that regenerate the paper tables and figures. Numerical conditions contain 397 entries plus country and index identifiers; the three streams are fused into the 128-dimensional condition vector described in Appendix A. The production seeds are 20260730–20260732 for repeated neural baselines and 20260730 for the frozen DLPM archive. The SHA-256 hashes of the untouched test CSV, frozen 40-path P archive, and unconditional-bootstrap archive are, respectively, `68B964021FC1D9F8...B8D8`, `7F44D27E0711F8F4...A974`, and `80C3EE618C497253...37B4`; complete hashes and environment requirements are stored in the release manifest. Raw market series remain subject to their providers' redistribution terms; the repository documents tickers, transforms, chronological boundaries, and archive checksums so authorized users can rebuild the experimental panel.

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